$$a_{\ell w} = \frac{m_w \exp g_\ell(\tilde{h}_w)}{\sum_{w'} m_{w'} \exp g_\ell(\tilde{h}_{w'})}, \qquad z_\ell = \left\langle \sum_w a_{\ell w}\, \tilde{h}_w,\ v_\ell \right\rangle + b_\ell$$